Solver Trace Analysis Reveals Pervasive Reward Hacking in GRPO-Trained Formal Reasoners

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Abstract

We introduce Solver Trace Analysis (STA), a diagnostic framework that applies leave-one-out (LOO) ablation to formal solver inputs to detect structural degeneration in GRPO-trained reasoning models. The derived metric, Inference Step Ratio (ISR), quantifies the fraction of input components that are logically necessary for proof success. Applying STA to Qwen3-4B trained with execution-only rewards on first-order logic (FOL) tasks, we find that all five random seeds exhibit $\geq$90% structural degeneration despite achieving high prove rates (mean 79%). We identify nine distinct reward hacking modes, including conclusion embedding (CE), tautological shortcuts (TS), and a meta-Goodhart phenomenon where the model simultaneously games execution, ISR, and hacking detection. When ISR is incorporated as a reward signal (β=0.1), evaluation across eight seeds reveals no statistically significant genuine reasoning advantage (Cochran–Mantel–Haenszel OR=1.28, p=0.14), indicating that ISR's structural bias toward tautologies creates an inherent Goodhart vulnerability at any dose below a model capacity threshold. Cross-logic validation on Prolog confirms universal degeneration (3/3 seeds, 0% genuine), while cross-scale analysis on Qwen3-1.7B reveals capacity-dependent hacking strategies. Our findings establish STA as a reliable diagnostic tool while demonstrating that structural reward signals face fundamental Goodhart limitations in the small-model regime.

1 Introduction

Group Relative Policy Optimization (GRPO; Shao et al. 2024) has emerged as a leading approach for training language models on formal reasoning tasks. By rewarding models based on solver execution outcomes—whether a generated proof is accepted by an automated verifier—GRPO achieves high prove rates without requiring human-annotated reasoning traces. However, high execution accuracy does not guarantee genuine reasoning: a model may learn to produce degenerate proofs that exploit solver behavior rather than constructing valid logical arguments.

This paper addresses the question: *how can we diagnose and characterize structural degeneration in* GRPO*-trained formal reasoners?* We propose **Solver Trace Analysis** (STA), a diagnostic framework that applies leave-one-out ablation to the premise set of each generated proof, re-executing the solver after removing each premise individually. The resulting metric, the **Inference Step Ratio** (ISR), measures the fraction of premises that are logically necessary for proof success. A genuine proof requires most of its premises (ISR $\approx$ 1.0), while a degenerate proof contains many redundant premises (ISR $\ll$ 1.0).

Applying STA to Qwen3-4B trained with execution-only GRPO on first-order logic tasks, we make three contributions:

C1: STA as Diagnostic. ISR detects reward hacking invisible to prove rate (PR). Across five seeds, exec-only GRPO achieves mean PR=79% but mean genuine rate of only 3.6%, with all seeds showing $\geq$90% structural degeneration. We identify nine distinct hacking modes organized into a taxonomy, revealing that degeneration mode is stochastically determined by random seed. At 1000 training steps, ISR collapse trajectories are mode-dependent: seeds locked into tautological shortcuts (TS) maintain ISR$\approx$0.5, while seeds transitioning to conclusion embedding (CE) collapse to ISR$\approx$0.

C2: Inevitable Goodhart Below Capacity Threshold. When ISR is used as a reward signal (β=0.1), expanded evaluation across N=8 seeds yields Cochran–Mantel–Haenszel (CMH) OR=1.28 (p=0.14), showing no statistically significant genuine advantage after controlling for seed-level heterogeneity. The initial apparent benefit (pooled OR=2.14) is driven by two outlier seeds, while 6/8 seeds show ISR-genuine rates at or below the random baseline. At β=0.05, 2/2 seeds exhibit deterministic TS collapse (97–100%). These results indicate that ISR's inherent structural bias—tautologies receive ISR$\geq$0.5—creates a Goodhart vulnerability at any dose for models below a capacity threshold.

C3: Cross-Logic Transfer. STA generalizes across formal languages. Prolog GRPO with structural rewards shows 3/3 seeds gaming with 0% genuine proofs, exhibiting logic-specific hacking strategies (degenerate syntax, query-as-fact insertion) distinct from FOL modes.

2 Related Work

RL for Formal Reasoning. Recent work applies RL to mathematical and formal reasoning (Shao et al., 2024; Luo et al., 2024; Wang et al., 2024). GRPO in particular uses group-relative advantages computed from multiple rollouts per prompt, avoiding the need for a separate critic model. While these methods achieve strong execution accuracy, the structural quality of generated solutions remains underexplored.

Reward Hacking and Goodhart's Law. Goodhart's law—that a measure ceases to be a good measure once it becomes a target—is well-documented in RL (Amodei et al., 2016; Pan et al., 2022; Skalse et al., 2022). In language model alignment, reward hacking manifests as models exploiting reward model weaknesses (Gao et al., 2023). Our work provides the first systematic empirical characterization of Goodhart's law in formal reasoning GRPO, documenting nine distinct hacking modes and demonstrating that structural metrics face the same vulnerability.

Proof Quality Metrics. Leave-one-out ablation for measuring component necessity has precedents in feature attribution (Covert et al., 2021) and minimal proof extraction. Our contribution is applying LOO ablation as a diagnostic for GRPO reward hacking in formal reasoning, and characterizing the failure mode when this diagnostic is itself used as a reward signal.

3 Method

3.1 Solver Trace Analysis

Given a generated proof consisting of a conclusion c and premise set $P = \{p_1, \ldots, p_n\}$, a formal solver $\mathcal{S}$ returns a binary judgment $\mathcal{S}(P, c) \in \{0, 1\}$. STA performs LOO ablation: for each $p_i \in P$, we evaluate $\mathcal{S}(P \setminus \{p_i\}, c)$. Premise p_i is *necessary* if $\mathcal{S}(P \setminus \{p_i\}, c) = 0$, i.e., removing it causes proof failure.

3.2 Inference Step Ratio

The Inference Step Ratio is defined as:

$$\text{ISR}(P, c) = \frac{|\{p_i \in P : \mathcal{S}(P \setminus \{p_i\}, c) = 0\}|}{|P|} \tag{1}$$

ISR $= 1.0$ indicates every premise is necessary (maximally tight proof); ISR $= 0$ indicates all premises are redundant. For a sample-level aggregate, ISR* denotes the mean ISR over all successfully proved samples in a batch.

Structural Bias. A critical property of ISR is that tautological proofs of the form $p \rightarrow p$ (a single premise restating the conclusion) receive ISR $= 1.0$ trivially—one premise, and it is "necessary" because removing it leaves an empty premise set. More generally, any proof with n premises where k are tautological copies of the conclusion receives ISR $\geq k/n$. This creates a positive feedback loop when ISR is used as a reward: tautologies are rewarded, reinforcing their generation.

3.3 Hacking Mode Taxonomy

Through manual inspection of 2,000+ generated proofs across conditions, we identify nine distinct reward hacking modes organized by mechanism:

1. **CE (Conclusion Embedding)**: The conclusion is embedded verbatim as a premise. Remaining premises are redundant. ISR $\approx 1/n$ where n is premise count.
2. **TS-a (Tautological Shortcut, simple)**: Proof reduces to $P \rightarrow P$. Exactly one or two components, ISR ≈ 0.5.
3. **TS-b (Tautological Shortcut, obfuscated)**: Tautology disguised through variable renaming or logical equivalence. Evades simple pattern matching. Represents *meta-Goodhart*: simultaneous gaming of execution reward, ISR reward, and hacking detection.
4. **PG (Premise Grafting)**: Unrelated premises conjoined with the conclusion via $\wedge$, creating artificial necessity. ISR inflated.
5. **CI (Contradiction Injection)**: Mutually contradictory premises make anything provable *ex falso*. ISR ≈ 0 (each contradiction partner alone suffices).
6. **PR Collapse**: Prove rate drops to near-zero; model generates ill-formed outputs. A degenerate failure mode rather than active gaming.
7. **Degenerate Syntax** (Prolog): Syntactically valid but semantically empty Prolog clauses.
8. **Query-as-Fact** (Prolog): The query is inserted as a database fact, trivially satisfying it.
9. **Fixed Template**: A single proof template used for all problems regardless of content (observed at 1.7B scale).

Modes 1–6 are observed in FOL; modes 7–8 are Prolog-specific; mode 9 is scale-specific (1.7B). A single seed typically converges to one dominant mode, though mode mixtures occur (Section 5.1).

4 Experimental Setup

Tasks. FOL tasks require generating a set of premises and a conclusion such that the premises logically entail the conclusion, verified by an automated theorem prover. Prolog tasks require generating clauses and a query such that SWI-Prolog succeeds on the query given the clauses.

Models. Primary: Qwen3-4B (4.0B parameters). Cross-scale: Qwen3-1.7B (1.7B parameters), Qwen3-8B (8.0B parameters, attempted).

Training. GRPO with K=8 rollouts per prompt, 500 training steps (extended to 1000 for trajectory analysis). Three reward configurations: (1) *exec-only*: binary solver execution reward; (2) ISR β: execution + $\beta \cdot$ ISR structural reward; (3) *random* β: execution + $\beta\cdot$ random structural noise (control). Seeds: s42–s46 for exec-only (5 seeds); s42–s49 for ISR β=0.1 (8 seeds).

Seed	PR	ISR*	Gen.%	Dominant Mode	Deg.%
s42	0.93	0.132	7.5	CE (87.1%)	92.5
s43	0.13	0.000	0.0	CE+Collapse	100.0
s44	0.98	0.582	0.0	TS (100%)	100.0
s45	0.99	0.506	1.0	TS (93.9%)	99.0
s46	0.93	0.083	9.7	CE (89.2%)	90.3
Mean	0.79	0.261	3.6	—	96.4

Table 1: Exec-only GRPO at 500 steps (Qwen3-4B, 5 seeds). All seeds exhibit $\geq$90% structural degeneration. PR = prove rate, Gen.% = fraction of proved samples classified as genuine, Deg.% = 100 − Gen.%. ISR* varies widely across seeds because different hacking modes have different inherent ISR values: TS ≈ 0.5 (tautology structure), CE ≈ 0.1 (redundant premises).

Seed	ISR^*_{100}	ISR^*_{500}	ISR^*_{1000}	Δ%	Mode at 1000
s42	0.642	0.132	0.003	−99.6	CE+PG
s43	—	0.000	0.082	—	CE (PR recovery)
s44	—	0.582	0.002	−99.7	CE+PG
s45	—	0.506	0.480	−5.1	TS (stable)
s46	—	0.083	0.065	−21.7	CE (deepening)

Table 2: ISR* trajectories from 500 to 1000 training steps. Collapse rate depends on hacking mode: TS-locked seeds maintain ISR*$\approx$0.5, while seeds transitioning to CE collapse >99%.

Evaluation. 100 held-out problems per evaluation. Each proof classified as *genuine* (valid reasoning with necessary premises) or assigned a hacking mode via automated classification validated by manual review. ISR* computed over all proved samples.

SFT Baseline. Supervised fine-tuning on gold-standard proofs: PR=14%, genuine=92.9%. Confirms that degeneration is specific to RL reward optimization, not task difficulty.

5 Results

5.1 Universal Degeneration Under Exec-Only Rewards

Table 1 summarizes results across five seeds at 500 training steps.

All five seeds show $\geq$90% structural degeneration, with mean genuine rate of only 3.6%. Critically, PR alone cannot distinguish genuine from degenerate: s44 achieves PR=98% with 0% genuine proofs (pure TS), while s46 achieves PR=93% with the highest genuine rate of 9.7%.

The five seeds exhibit four distinct hacking mode distributions: s42 and s46 are CE-dominant, s44 and s45 are TS-dominant, and s43 shows CE with prove-rate collapse. This *stochastic bifurcation*—where the dominant hacking mode is determined by random seed—implies that single-seed evaluations of GRPO structural quality are unreliable.

ISR* ranges from 0.000 (s43) to 0.582 (s44), a 50$\times$ spread driven by hacking mode: TS proofs structurally contain $\sim$1 necessary component out of 2 (ISR ≈ 0.5), while CE proofs embed the conclusion among redundant premises (ISR $\ll 0.5$). This means ISR* must be interpreted jointly with hacking mode; its absolute value alone does not indicate degeneration severity.

5.2 Mode-Dependent ISR Collapse Trajectories

Table 2 extends the exec-only analysis to 1000 training steps. The key finding is that ISR collapse is *mode-contingent*, not universal:

Statistic	Value	Interpretation
Pooled OR	2.14	Apparent ISR advantage
CMH OR	1.28	No significant advantage
CMH p-value	0.14	n.s. at α=0.05
Breslow–Day p	$<$0.001	Significant heterogeneity
I^2	89%	Very high inconsistency
Genuine window	2/8 (25%)	Minority of seeds benefit

Table 3: Statistical summary for ISR β=0.1 vs. random β=0.1 across N=8 seeds. The apparent pooled advantage (OR=2.14) is driven by two outlier seeds (s45, s46) and disappears after seed stratification (CMH OR=1.28, p=0.14).

- **TS-locked seeds (s45)**: ISR* remains at $\approx$0.5 because the tautological proof structure $P \to P$ has an inherent lower bound of 0.5 (one necessary component out of two). Additional training refines but does not escape this structural fixed point.
- **CE-transitioning seeds (s42, s44)**: ISR* collapses $>$99% as the model converges to pure conclusion embedding with all other premises redundant.
- **CE-stable seeds (s46)**: ISR* declines slowly (0.083 $\to$ 0.065) as CE purity increases from 89% to 94%.
- **Recovering seeds (s43)**: ISR* increases from 0.000 to 0.082 as the model recovers from PR collapse, though into CE-dominated degeneration rather than genuine reasoning.

The trajectory for s42—ISR* declining from 0.642 (100 steps) to 0.132 (500 steps) to 0.003 (1000 steps), a 99.6% collapse—provides the clearest illustration of progressive structural degeneration hidden behind stable prove rates.

5.3 ISR as Reward: No Robust Genuine Advantage

We evaluate ISR as a reward signal at β=0.1 across N=8 seeds (s42–s49), with matched random-noise controls. Table 3 presents the statistical summary.

The pooled odds ratio of 2.14 suggests an ISR advantage, but this is misleading. The Cochran–Mantel–Haenszel test, which controls for seed-level stratification, yields OR=1.28 (p=0.14)—not significant. The Breslow–Day test confirms significant heterogeneity (p $<$0.001), and I^2=89% indicates that 89% of variance is due to between-seed differences rather than sampling noise.

Table 4 reveals the source of heterogeneity: seeds s45 and s46 show strong ISR advantages (OR=13.0 and 56.7), while the remaining six seeds show ISR genuine rates at or below the random baseline.

This pattern—a small minority of "lucky" seeds escaping Goodhart collapse while the majority succumb—replaces the earlier bimodal narrative with a simpler characterization: ISR *as a reward signal occasionally produces genuine reasoning by chance, but this effect is not statistically reliable.*

No Safe Dose. At β=0.05, both tested seeds (s42, s44) exhibit deterministic TS collapse (97–100% TS), eliminating the possibility that a lower dose avoids Goodhart. At β=0.3, three seeds show multi-modal instability with PR=66$\pm$42%, confirming non-monotonic dose-response: β=0.05 (deterministic TS) $\to$ β=0.1 (seed-contingent) $\to$ β=0.3 (multi-modal collapse). The ISR structural bias—tautologies receiving ISR$\geq$0.5—is the mechanistic root: this creates an optimization attractor that dominates the reward landscape regardless of β.

Meta-Goodhart. Seed s44 under ISR β=0.1 produces obfuscated tautologies (TS-b): proofs that are structurally tautological but disguised through variable renaming, simultaneously gaming execution reward, ISR reward, and hacking detection. This *meta-Goodhart triple gaming* represents the most so-

Seed	ISR		Random		OR	Dir.
	N_{pv}	Gen%	N_{pv}	Gen%		
s42	81	17.3	79	22.8	0.71	Rand↑
s43	61	0.0	94	1.1	0.00	Rand↑
s44	89	0.0	89	6.7	0.00	Rand↑
s45	42	54.8	58	8.6	13.0	ISR↑↑
s46	77	54.5	94	2.1	56.7	ISR↑↑
s47	97	3.1	89	6.7	0.44	Rand↑
s48	66	0.0	93	4.3	0.00	Rand↑
s49	86	1.2	5	0.0	—	N/A
Pool	599	13.9	601	7.0	2.14	ISR↑

Table 4: Per-seed genuine rates for ISR β=0.1 vs. random β=0.1 (N=8). N_{pv} = number of proved samples. Only 2/8 seeds (s45, s46) show ISR advantage; the remaining 6 show the opposite direction. s49-random collapsed to N_{pv}=5 (repetition failure).

Seed	Raw ISR*	Matched ISR*	Δ	$\lvert P\rvert_{\text{hack}}$
s42	0.132	0.118	-0.014	3.2
s43	0.000	0.000	0.000	18.9
s44	0.582	0.495	-0.087	2.0
s45	0.506	0.492	-0.014	2.1
s46	0.083	0.071	-0.012	4.1
Mean	0.261	0.235	-0.025	—

Table 5: Effect of premise-count matching on ISR* (exec-only). Raw ISR* is corrected by -8.5% on average when controlling for the number of premises. $\lvert P\rvert_{\text{hack}}$ = mean premise count for hacking samples.

phisticated hacking mode observed, where the model optimizes against three objectives, including the diagnostic itself.

5.4 Premise-Count Confound

LOO ablation metrics like ISR are confounded by proof complexity: proofs with more premises mechanically have more opportunities for individual premises to appear necessary. We quantify this by computing premise-count-matched ISR*, comparing genuine and hacking proofs only within the same premise-count stratum.

Table 5 shows that matching reduces ISR* by 8.5% on average. The correction is largest for TS-dominant seeds (s44: Δ=-0.087) where hacking proofs have very few premises ($|P|$=2.0) compared to genuine proofs (mean $|P|$=7.2). Genuine proofs have significantly more premises than hacking proofs overall (7.2 vs. 3.8, p=0.03), confirming the confound.

This is a methodological contribution: LOO ablation metrics are widely used in interpretability and proof analysis without controlling for component count. Documenting this confound provides actionable guidance for future work. Importantly, matched ISR* retains diagnostic utility—genuine proofs still achieve substantially higher matched ISR* ($\approx$0.45) than hacking proofs ($\approx$0.08)—so the confound modulates but does not invalidate the diagnostic.

5.5 Cross-Scale and Cross-Logic Validation

Cross-Scale. Table 6 presents results across model scales.

At 1.7B scale, Qwen3-1.7B with random β=0.1 achieves PR=93% but 100% CI (contradiction injection) mode, generating mutually contradictory premises that make anything provable. With ISR β=0.1, the model achieves PR=100% with ISR=0.6 (zero variance) via a single fixed proof template

Scale	Gen.%	Dominant Mode	Notes
1.7B	0%	CE/CI (fixed template)	Template gaming
4B	3.6%	CE/TS (seed-dep.)	Stochastic bifurcation
8B	N/A	—	Training failure

Table 6: Cross-scale exec-only results. 1.7B and 4B both show near-total degeneration but with scale-dependent hacking strategies. 8B training failed due to VRAM constraints at K=8 and cold-start stagnation at K=4.

(CE mode)—the simplest possible exploit of the combined reward. At β=0.3, learning suppression occurs (PR=17%), with the 29.4% genuine rate attributable to the model's inability to optimize at all rather than genuine skill.

These results establish *capacity-dependent Goodhart basins*: 1.7B converges to the simplest exploits (fixed templates, contradictions), while 4B discovers more sophisticated strategies (TS, obfuscated tautologies). The hacking strategy scales with model capacity, but hacking itself is universal below a capacity threshold.

Qwen3-8B training was attempted but did not yield results. At K=8, estimated VRAM requirements ($\sim$112GB) exceeded the available 98GB. At K=4, cold-start stagnation occurred (PR$<$5% after 500 steps), as the reduced rollout count provided insufficient variance for GRPO's group-relative advantages. This limitation prevents us from determining whether a capacity threshold exists above which Goodhart collapse is avoided, a question we leave to future work with larger computational budgets.

Cross-Logic (Prolog). Prolog GRPO with structural rewards confirms universal Goodhart:

- **Exec-only** (3 seeds): PR=53.3%$\pm$31.1%, ISR near-zero (2/3 seeds $<$0.04).
- **Structural reward** (3 seeds): 3/3 seeds gaming, 0% genuine. Seed-specific modes: s42 = clean TS (100%), s43 = degenerate syntax (97.4%), s44 = query-as-fact (97.9%).

Prolog degeneration is more extreme than FOL (0% vs. $\sim$3.6% genuine residual), consistent with Prolog's simpler semantics offering more complete exploitation paths. The emergence of logic-specific hacking modes (degenerate syntax, query-as-fact) that have no FOL analog confirms that reward hacking is not a fixed behavioral pattern but an adaptive optimization response to the reward landscape.

6 Discussion

Diagnostic vs. Optimization. Our results draw a sharp distinction between ISR as a diagnostic tool and ISR as an optimization target. As a diagnostic, ISR reliably detects structural degeneration invisible to prove rate, across logic types, model scales, and training durations. As a reward signal, ISR suffers from the same Goodhart vulnerability it aims to diagnose, because tautological proofs receive inherently high ISR scores. This asymmetry—reliable for measurement, unreliable for optimization—is the central empirical finding of this paper.

Stochastic Bifurcation. The dominant hacking mode is determined stochastically by random seed. Five seeds produce four distinct mode distributions; eight seeds show only 2/8 escaping Goodhart collapse. This has practical implications: any evaluation of GRPO structural quality based on fewer than five seeds risks reporting seed-specific artifacts as general phenomena.

Mode-Dependent Diagnostics. ISR* values must be interpreted jointly with hacking mode. A seed with ISR*=0.5 may be entirely degenerate (TS mode) rather than partially genuine. Future diagnostics should report mode distributions alongside aggregate metrics.

Capacity Threshold Hypothesis. Our results are consistent with a capacity threshold below which Goodhart collapse is inevitable: both 1.7B and 4B exhibit near-total degeneration, while 8B results remain unknown due to training failure. Whether larger models can sustain genuine reasoning under execution-only rewards is an open question with significant implications for scaling formal reasoning with RL.

7 Conclusion

We introduced Solver Trace Analysis (STA), a diagnostic framework for detecting reward hacking in GRPO-trained formal reasoners. Applied to Qwen3-4B on first-order logic, STA reveals that high prove rates mask pervasive structural degeneration: all five exec-only seeds show $\geq$90% degeneration across nine distinct hacking modes. Expanded evaluation (N=8) demonstrates that ISR as a reward signal provides no statistically robust genuine advantage (CMH OR=1.28, p=0.14), establishing that structural reward signals face fundamental Goodhart limitations below a model capacity threshold. Cross-logic validation on Prolog (3/3 gaming, 0% genuine) and cross-scale analysis at 1.7B confirm the generality of these findings. The nine-mode hacking taxonomy, meta-Goodhart phenomenon, and premise-count methodology provide actionable diagnostic tools for the community.

Limitations

Our study has three main limitations. First, the cross-scale evidence is incomplete: Qwen3-8B training failed due to computational constraints, leaving the capacity threshold hypothesis untested at larger scales. Second, all experiments use a single model family (Qwen3); generalization to other architectures remains unverified, though the Goodhart mechanism (ISR's inherent tautology bias) is architecture-independent. Third, ISR measures syntactic necessity rather than semantic faithfulness—premises that are syntactically necessary may still be semantically vacuous—bounding STA's diagnostic scope to structural quality.

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